

Research Paper

A comparative study of machine learning approaches for an accurate predictive modeling of solar energy generation

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ABSTRACT

Solar energy prediction poses a challenging task that necessitates robust models and precise data to accurately forecast solar energy yield, especially in grid areas with high photovoltaic shares. Existing methods often rely on statistical or physical models, which have limitations in capturing the complex and non-linear relationships between weather variables and solar power generation. In this paper, we address this issue by comparing and evaluating different learning models, ranging from artificial neural networks (ANNs) and random forest models to long- and short-term memory (LSTM) networks, to predict the PV energy yield based on weather forecast data. A methodology has been developed to evaluate various models using real-world datasets from a large-scale industrial solar project, incorporating historical photovoltaic data, meteorological data, and solar irradiation data. The experimental results showed that the Random Forest Algorithm (RFR) consistently outperforms other algorithms, providing a mean absolute error (MAE) of 0.06 and a root mean square error (RMSE) of 0.15 when applied to historical meteorological datasets. The accuracy of the learning model was improved by combining meteorological data with a solar irradiation dataset to obtain an MAE of 0.03 % and an RMSE of 0.09 %. Validation analysis showed the proposed model is effective in both forecast accuracy and stability. The proposed methodology can provide valuable information to PV system operators, grid managers, and energy planners, facilitating the optimization of solar energy resources.

1. Introduction

As the demand for renewable energy continues to increase, the efficient utilization of solar power plays a crucial role in sustainable energy solutions. Integrating a higher proportion of renewable sources, such as solar power, is advantageous to reduce carbon emissions and meet future power grid requirements. However, this integration also poses new challenges related to grid management. Variability and uncertainty in photovoltaic (PV) energy generation can lead to stability and reliability issues in power system operations due to the intermittent nature of solar generated electricity. Consequently, grid operators must incorporate these factors into their generation planning and dispatch operations (Mahoor et al., 2018; Sadeghian and Wang, 2020). This has led utility grids with numerous distributed photovoltaic systems to transition to modern and digitally enhanced technologies. These technologies facilitate the monitoring and control of distributed energy resources. The ongoing trends of electrification, decentralization, and digitalization are driving the transformation of the current paradigm of

the power sector. This transformation aims to fully exploit the flexibility of the system to accommodate elevated levels of variable renewable energy

Saudi Arabia has set ambitious goals to generate 50 % of its energy from renewable sources by 2030 (Sekhar, 2020). To achieve this goal, the country has actively invested in renewable energy initiatives, focusing on solar energy as a key component of its renewable energy strategy. However, the integration of PV systems into the energy grid presents challenges due to the variability of solar power generation, which is influenced by weather and environmental conditions (Chuluunsai Khan et al., 2021; Almonacid-Ollerios et al., 2020). Accurate forecasting of solar power production is crucial for effective energy management and grid stability (Almonacid-Ollerios et al., 2020).

Several research investigations have explored the impact of adverse weather conditions, such as wind speed/direction, temperature, relative humidity, and frequency of dust storms, on the losses in the power output of photovoltaic cells. Understanding the dynamics of solar panels is crucial to ensure proper installation and achieve optimal performance

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(Almonacid-Ollerios et al., 2020; Sayyah et al., 2014).

The use of artificial neural networks for the prediction of solar power has been addressed in many recent works. In (Liu et al., 2017a), a backpropagation neural network is used to forecast solar power 24 hours ahead. Similarly, the Elman neural network is used in (Jayawardene and Venayagamoorthy, 2014) to achieve a 72-hour photovoltaic power forecast. In (Mandal et al., 2013), the authors integrate Wavelet Transform (WT) with a generalized regression neural network for forecasting short-term PV power. Fuzzy neural networks (FNN) are suggested for the estimation of PV power in another study (Wang et al., 2012).

Another approach involves ensemble learning algorithms. In (Jayawardene and Venayagamoorthy, 2014), the authors apply an extreme learning machine algorithm for the multistep prediction of photovoltaic power. In (Lahouar et al., 2017), the random forest algorithm is utilized by the authors for forecasting the day-ahead of solar power. In (da Silva Fonseca et al., 2012), the support vector regression (SVR) is used to estimate the PV power output, examining the impact of cloudiness on forecast accuracy. Furthermore, in (Wolff et al., 2016), the authors compare the K-Nearest-Neighbors (KNN) and SVR methods using actual measurements and Numerical Weather Prediction (NWP) data, employing feature extraction to identify the top ten statistics for model input.

A literature review reveals that machine learning methods typically outperform conventional statistical approaches. This superiority is primarily attributed to their capacity to handle non-linear and dynamic models. However, the process of training artificial neural networks continues to be a challenging task. There are no one-size fits-all approach, and it often involves a lot of trial and error. Many studies also only explore a few machine learning methods and lack details on how the model settings were chosen. Additionally, these studies are often done in different real-world settings, making it difficult to compare them directly.

The main objective of this paper is to determine a robust and accurate PV output energy prediction methodology by evaluating different training methods for a real-world case study. To do this, a comparative study of widely used machine learning methods such as artificial neural networks (ANN), Long Short-Term Memory (LSTM), K-nearest neighbors (KNN), and random forest regression (RFR) was performed to evaluate their effectiveness for PV power forecasting applications.

The proposed methodology included training machine learning models in different supervised learning scenarios and benchmarking forecast performance. The verification was carried out using actual high-quality hourly PV operational and meteorological measurements acquired over two years from a large-scale industrial photovoltaic plant in Saudi Arabia.¹ The settings of the test bench photovoltaic system provide a perfect opportunity to study the effect of supervised training regimes on the accuracy of forecasts based on commonly used metrics. Furthermore, the results obtained provided useful information for the establishment of a robust forecasting methodology that combines historical data and weather forecasts with an optimal supervised learning approach.

The primary contributions of the paper encompass the following key points:

- Establish a resilient PV power forecasting system through a comprehensive evaluation of various machine learning algorithms, thereby identifying the most precise model.
- Elevating the precision of the prediction of the PV power output by incorporating weather-related features alongside historical data.
- Enhancing the efficiency and accuracy of the performance and forecasting of the network connection.

The remainder of this paper is structured as follows: In Section 2, we provide a review of the literature and discuss related work. Section 3 presents the case study, including its datasets and the feature engineering process. Section 4 describes the methodology used to evaluate and compare various learning algorithms. Section 5 presents the experimental setup and presents the results of our work. Finally, in Section 6, we conclude and highlight key perspectives from this study.

2. Related works

In recent years, there has been a significant increase in attention and advancement in research related to solar energy prediction. Numerous studies have focused specifically on developing accurate prediction models using machine learning techniques. An approach commonly adopted in solar power forecasting is the use of artificial neural networks (ANNs) (Da Silva et al., 2017). ANNs are effective in capturing the non-linear relationships between weather variables and solar power generation (Saberian et al., 2014; Lee and Kim, 2019; Ozoegwu, 2019; Ozdemir et al., 2022). In (Saberian et al., 2014), Saberian et al. utilized artificial neural networks to forecast the output power of solar panels based on meteorological conditions including temperature and humidity. Similarly, Lee and Kim (2019) explored three methods, namely ANNs, deep neural networks (DNN), and long- and short-term memory (LSTM) models, to predict PV power outputs by capturing hidden relationships within meteorological data. In (Ozoegwu, 2019), the authors highlight the effectiveness of artificial neural networks in forecasting solar radiation. They emphasize the importance of incorporating time-series data and month-specific patterns, which contribute to improving the accuracy of solar energy predictions. In (Ozdemir et al., 2022), the authors used artificial neural networks (ANN) and long short-term memory (LSTM) to predict solar radiation. The objective of this paper is to help harness solar energy more efficiently, especially in regions with intermittent electrical power. The authors evaluated the experimental results by comparing their obtained results using metrics such as the coefficient of determination (R^2), mean square error (MSE), root mean squared error (RMSE), mean absolute error (MAE), and mean biased error (MBE).

Another popular machine learning technique for solar power forecasting is support vector regression (SVR), which has been applied in numerous studies to predict solar power production based on weather data (Nageem and Jayabarathi, 2017; Lee and Kim, 2019; Kabilan et al., 2021). In (Nageem and Jayabarathi, 2017), Nageem et al. introduced a multi-input support vector regression (SVR) model to forecast the performance of solar panels connected to the grid. In their model, humidity, temperature, pressure, and wind speed are incorporated as input variables. In (Lee and Kim, 2019), the authors used the HIMVO-SVM model to predict PV output power in different weather scenarios using historical data provided by a solar center in Australia. The results demonstrated the effectiveness of the SVR model in accurately predicting solar power output, particularly in capturing the non-linear relationships between input features and output predictions. In (Kabilan et al., 2021), the authors employ various machine learning algorithms, including support vector regression, to predict the power output of building integrated photovoltaic systems. The study aims to determine the most accurate and reliable algorithm for short-term power prediction.

Ensemble methods, such as Random Forest Regression (RFR), have also been extensively explored in solar power forecasting. RFR combines multiple decision trees to make predictions and has been shown to achieve high accuracy and robustness (Czajkowski and Kretowski, 2016). In (Villegas-Mier et al., 2022), the authors present a methodology to forecast the solar power output using a random forest algorithm. The methodology is based on a dataset of historical solar power output and weather data, including solar radiation, temperature, and wind speed. The authors applied their methodology to a solar power production dataset in China and compared the results with those obtained using other forecasting methods. The results showed that the RFR algorithm

¹ Sakaka PV IPP, ACWA POWER, <https://acwapower.com/en/projects/sakaka-pv-ipp/>

Table 1
Related works summary.

Algorithm	Variables	Metrics	References
ANN	Weather and air pollution including temperature, humidity, wind speed, and solar radiation	MAE: 0.35	(Chuluunsaikhan et al., 2021)
		RMSE: 0.85	
		RMSE: 0.055	(Lee and Kim, 2019)
		MAE: 0.038	
		MAE:0.051	
		RMSE: 0.063	
		MAE: 0.044	(Theocharides et al., 2020)
		MAPE: 5.71 %	
		MAE: 0.17	(Ozdemir et al., 2022)
		RMSE: 0.089	
RFR	Solar radiation, Temperature, Wind speed	MAE: 0.042	(Liu et al., 2019)
		MAPE: 4.65 %	(Zhang et al., 2018)
		MAE: 0.035	
		MAPE: 18.56 %	(Lee and Kim, 2019)
		MAE: 0.055	
SVR	Month, hour, Diffuse Horizontal Irradiance, Temperature, humidity	RMSE: 0.066	(Nageem and Jayabarathi, 2017)
		MAPE: 36 %	
		MAE: 0.062	(Lee and Kim, 2019)
		RMSE: 0.073	
		MAE: 8.87	
HIMVO-SVM	Solar radiation intensity, humidity, atmospheric temperature	MAE: 0.026	(Kabilan et al., 2021)
LSTM	PV power output data. Meteorological parameters include temperature, humidity, solar radiation, wind speed, etc.	MAPE: 1.87 %	(Li et al., 2019)
		RMSE: 0.8619	
		MAE: 0.05	(Kumari and Toshniwal, 2021; Guanoluisa et al., 2023)
		MAE: 2.86	
		MAPE: 9.65 %	
Hybrid Model	Temperature, humidity, pressure, wind speed, direction, rainfall, snowfall, snow depth	MAE: 0.273	(Sharma et al., 2022)
		MSE: 0.2877	
		R ² : 92.7 %	(Su et al., 2019)
		nRMSE: 6.74 %	
		MAE: 0.032	
		MAPE: 13.1 %	(Theocharides et al., 2020)
		MAE: 0.029	
		MAPE: 3.13 %	(Liu et al., 2019)

outperforms the other methods, achieving a mean absolute error (MAE) of 3.58 and a mean absolute percentage error (MAPE) of 18.56 %.

Furthermore, from sequential and time-series data, recurrent neural networks (RNNs) can learn features and long-term dependencies (Salehinejad et al., 2017). An example of RNN is Long Short-Term Memory (LSTM) networks, which have gained popularity in solar power forecasting due to their ability to capture temporal dependencies in time series data (Kumari and Toshniwal, 2021; Lee et al., 2018). In (Kumari and Toshniwal, 2021), an LSTM-based model was proposed to predict solar power generation based on historical weather data. The results showed that the LSTM model achieved superior performance compared to traditional machine learning models, highlighting the importance of considering temporal dynamics in solar power forecasting. In (Lee et al., 2018), a hybrid model that combined LSTM with a particle swarm optimization algorithm achieved enhanced accuracy in solar power forecasting. Deep learning models, including LSTM and CNN, are often considered black-box models, which means that it can be challenging to interpret the inner workings and understand the reasons behind their predictions. Providing insights into the interpretability of the proposed model could improve its practical utility and acceptance in real-world applications.

In addition, hybrid models that combine multiple forecasting techniques have also been investigated. These models aim to leverage the strengths of different algorithms to improve prediction accuracy (Su et al., 2019; Sharma et al., 2022; Theocharides et al., 2020; Liu et al., 2019). In (Su et al., 2019), Di Su et al. adopted a hybrid approach that combined multiple uncorrelated forecasting techniques to improve the accuracy of their predictions. Note that none of the recent studies in the literature have specifically addressed the challenges of long-term projections for such large-scale solar plants. In (Sharma et al., 2022), Jatin et al. tested three optimizers and found that the Nadam optimizer outperformed ARIMA and SARIMA, suggesting its potential applicability even to smaller plants with reduced parameter measurements while maintaining high accuracy. In (Theocharides et al., 2020), the authors address the complex nature of PV power forecasting by integrating machine learning and statistical post-processing techniques. This hybrid approach leverages the strengths of both methods, allowing for more accurate and reliable predictions. In (Liu et al., 2019), the authors of the article propose a hybrid model that combines artificial neural networks and support vector regression for solar power forecasting. The hybrid model is designed to take advantage of both techniques to improve the accuracy of forecasting for solar power generation. The article lacks a detailed analysis of the model’s performance compared to existing methods or benchmarks. The study evaluates machine learning models to improve day-ahead forecasting for PV power generation, with BNN proving to be highly effective with an nRMSE of 4.53 % (Theocharides et al., 2021). A study by Richard et al. compares deep learning models for accurately forecasting photovoltaic power generation in Ecuador’s Galapagos Islands. In particular, the LSTM model achieved a low RMSE

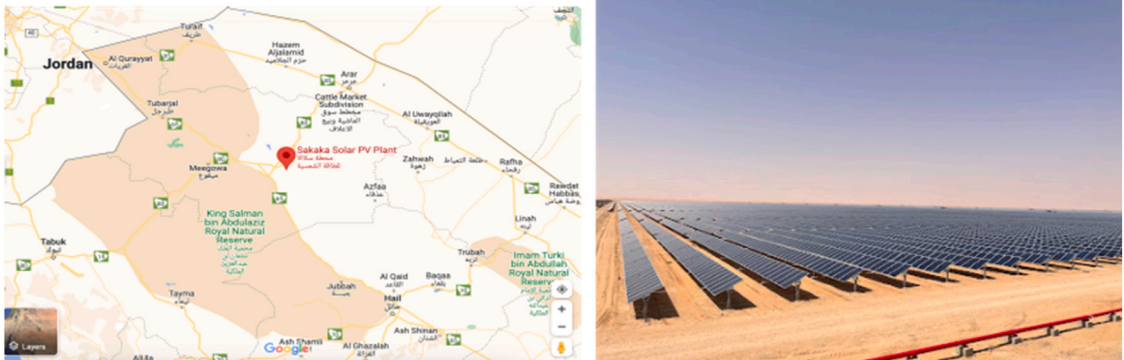


Fig. 1. IPP PV plant, Sakaka, Saudi Arabia.

Table 2

Solar irradiance data.

Name	Description	Unit	Type
DateTime	Timestamp	Time	DateTime
WMS i/POA1 (i=1...5)	Irradiation module 2	Watt/ m ²	float64
WMS i/POA2 (i=1...5)	Irradiation module 2	Watt/ m ²	float64
WMS i/Module1Temp (i=1...5)	Module 1 temperatures	Watt/ m ²	float64
WMS i/Module2 Temp (i=1...5)	Module 2 temperatures	Watt/ m ²	float64
Pyranometer i (i=1...10)	Solar radiation	Watt/ m ²	float64

of 0.05, highlighting its effectiveness in short-term prediction for sustainable energy planning (Guanoluisa et al., 2023). Table 1 summarizes the related works, the relevant variables used, and the performance obtained.

3. Case study and data engineering

Data for this study was obtained from a commercial solar photovoltaic plant in Sakaka operated by SAKAKA Solar Energy Company (Fig. 1). Sakaka is a city in northwest Saudi Arabia that is the capital of the Al Jawf province. The climate in Sakaka is a tropical and subtropical desert climate. The average temperature for the year is 22°C. The hottest month is July with an average temperature of 32°C. The month of January is the most humid with an average relative humidity of 57 %.

The plant has a capacity of 300 MW, the first utility-scale renewable energy project in Saudi Arabia under the country's National Renewable Energy Program. With a budget of 302 million USD, this project is the first in a series of projects under Saudi Arabia's national renewable energy program, which aims to achieve 50 % renewable energy generation in Saudi Arabia by 2030 (Sekhar, 2020). The Sakaka PV IPP plant began commercial operation at the end of 2019. We used two years of data from July 2020 to June 2022.

Solar irradiance data play a crucial role in the analysis and optimization of solar energy systems. In this study, comprehensive solar irradiance data were collected for the entire plant and stored in the data file 'Sakaka WMS.csv'. The dataset consists of three types of variables: pyranometer global horizontal irradiance, POA (Plane of Array) solar irradiance, and module temperature.

To ensure accurate measurements, ten pyranometers were strategically placed to cover the entire surface area of the solar plant (Pyranometer1 to Pyranometer10). These pyranometers measure the total amount of solar radiation that reaches the Earth's surface horizontally. Furthermore, ten POA modules, WMS₁/POA1 to WMS₅/POA1 and WMS₁/POA2 to WMS₅/POA2, were used to measure solar radiation that falls directly on solar panels. These measurements consider several factors, such as the angle of the Sun, atmospheric effects on sunlight, and the shading caused by surrounding objects. POA solar irradiance is a critical factor in determining the efficiency and power output of a solar panel system. The plant temperature is measured using ten modules: WMS₁/Module1 Temp to WMS₅/Module1 Temp and WMS₁/Module2 Temp to WMS₅/Module1 Temp. Solar irradiance data was collected over a two-year period, from July 2020 to June 2022. Table 2 provides a summary of the features of the dataset, their descriptions, units, and encoding.

In general, little maintenance was required of the instruments during the campaign. The pyranometers partially malfunctioned for two periods between 10 June 2020 and 13 June 2020 and between 14 July and 18 July 2021 and the corresponding data are omitted. To reduce the feature space and select the most relevant variables, we used Pearson's correlation to identify dependencies between variables. We identified an extremely high correlation of 99 % between the values of the

Table 3

Solar data frame after feature selection.

Name	Description	Unit	Type
DateTime	Hourly timestamp	Time	DateTime
POAAvg	Mean value of plane of array (POA) modules	Watt/m ²	float64
TempAvg	Mean value of temperature modules	°C	float64
IrradiationAvg	Mean value of pyranometers	Watt/m ²	float64

temperature modules, pyranometers, and POA modules, and this is for both modules. Therefore, we computed an average value for each of these three physical parameters and added it to the dataset while removing the original columns from the dataset to simplify it. The DateTime column is used as the index of the dataset. Furthermore, due to an error in the system timestamping, the DateTime has been shifted by 12 hours. All negative values in the dataset have been eliminated and outliers have been filtered using the z-score method. The resulting dataset has the structure shown in Table 3. Fig. 2 presents a sample of the dataset and the main statistics of the variables.

The second data file 'Sakaka Energy Data.csv' contains the cumulative solar power data per hour exported by two modules PT1 and PT2, and they are in kilowatts (Kwh) and megawatts (Mwh) for the period from July 2020 to June 2022.

Fig. 3 presents a sample of the solar power dataset with the main statistics of the different variables. It has been observed that there exists a flawless correlation between PT1 and PT2, as shown in Fig. 4. Consequently, only the PT1 column is retained for subsequent phases. The hourly output energy of the PT1 module is calculated from the cumulative value of PT1. The structure of the second data set after data engineering is depicted in Table 4.

The third data file 'Sakaka Weather Energy Data.csv' combines PV system power and weather type for the period from November 2020 to July 2022 as described in Table 5.

The type of weather is a categorical variable with the following values: cloudy, hazy, sunny, and mostly sunny. The file does not include the night data since the PV system does not produce any power at night. The times the system starts and stops are given to show its operation.

After collecting, cleaning, and merging the data, the attributes needed to forecast the PV power output are determined. To determine the importance of each feature in predicting PV power output, a Principal Component Analysis (PCA) is performed. The analysis revealed that the three most significant characteristics are irradiance, temperature, and type of weather.

Fig. 5 describes the correlation between the different features used and selected for the learning algorithms.

4. Proposed methodology

This section describes the methodology proposed to evaluate and compare the different learning algorithms as described in Fig. 6.

4.1. Data collection

Data are collected from various sources, such as Sakaka Weather Monitoring System (Sakaka WMS.csv), Sakaka energy data (Sakaka Energy Data.csv), and Sakaka weather data (Sakaka Weather Energy Data.csv). These datasets provide the information necessary to train and test machine learning models.

4.2. Data wrangling

The collected data undergoes several preprocessing steps, including cleaning, normalization, resampling, and aggregation. These steps ensure that the data are in a suitable format and ready for further analysis.

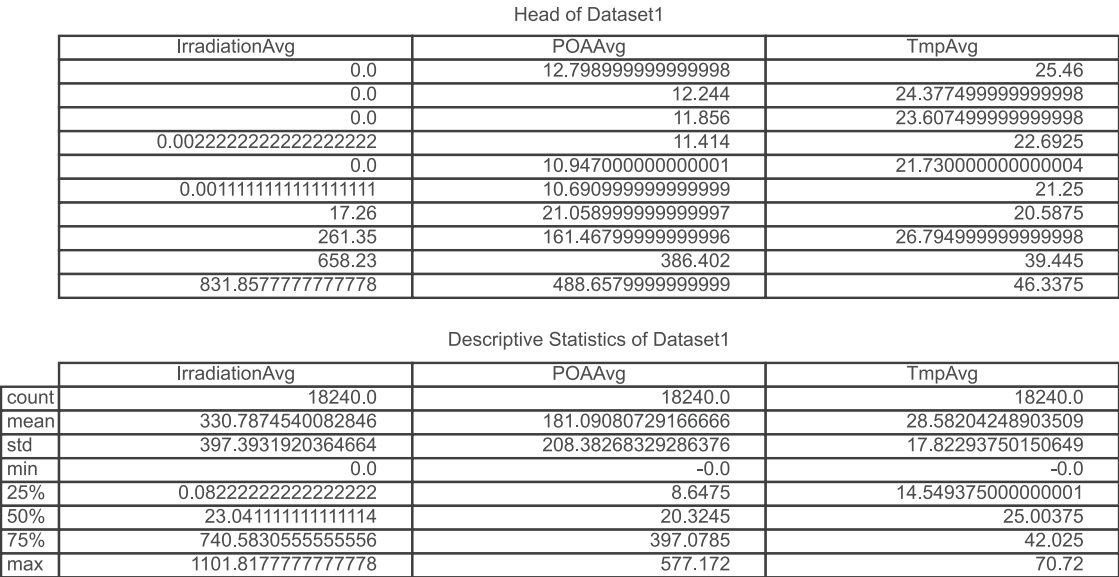


Fig. 2. Descriptive statistics of the solar irradiance dataset.

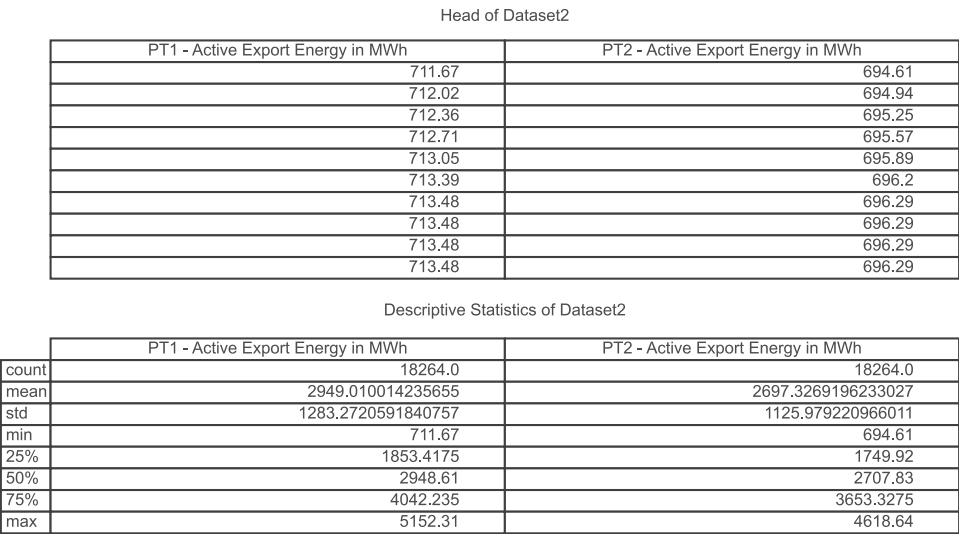


Fig. 3. Descriptive statistics of the solar power dataset.

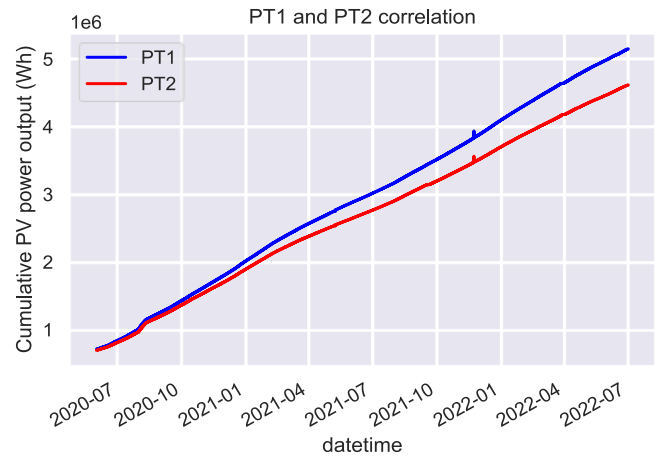


Fig. 4. Correlation between PT1 and PT2.

Table 4
Cumulative solar power output per hour for PT1.

Name	Description	Unit	Type
DateTime	Hourly timestamp	Time	DateTime
HourlyEnergy	Hourly solar energy for PT1	KW (KiloWatt)	float64

Table 5
PV system power and weather type.

Name	Description	Unit	Type
DateTime	Hourly timestamp	Time	DateTime
Startup time	System startup	Time	DateTime
Shutdown time	System shutdown	Time	DateTime
Energy	PV system power per hour	KW (KiloWatt)	float64
Weather	Weather type: Sunny, cloudy, etc.	Text	Categorical

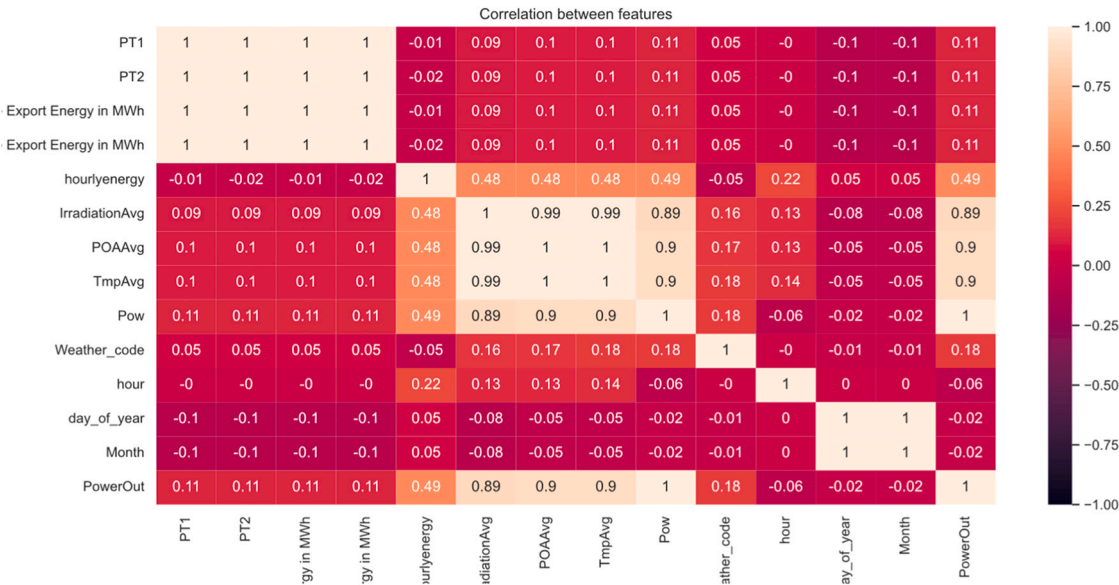


Fig. 5. Features correlation.

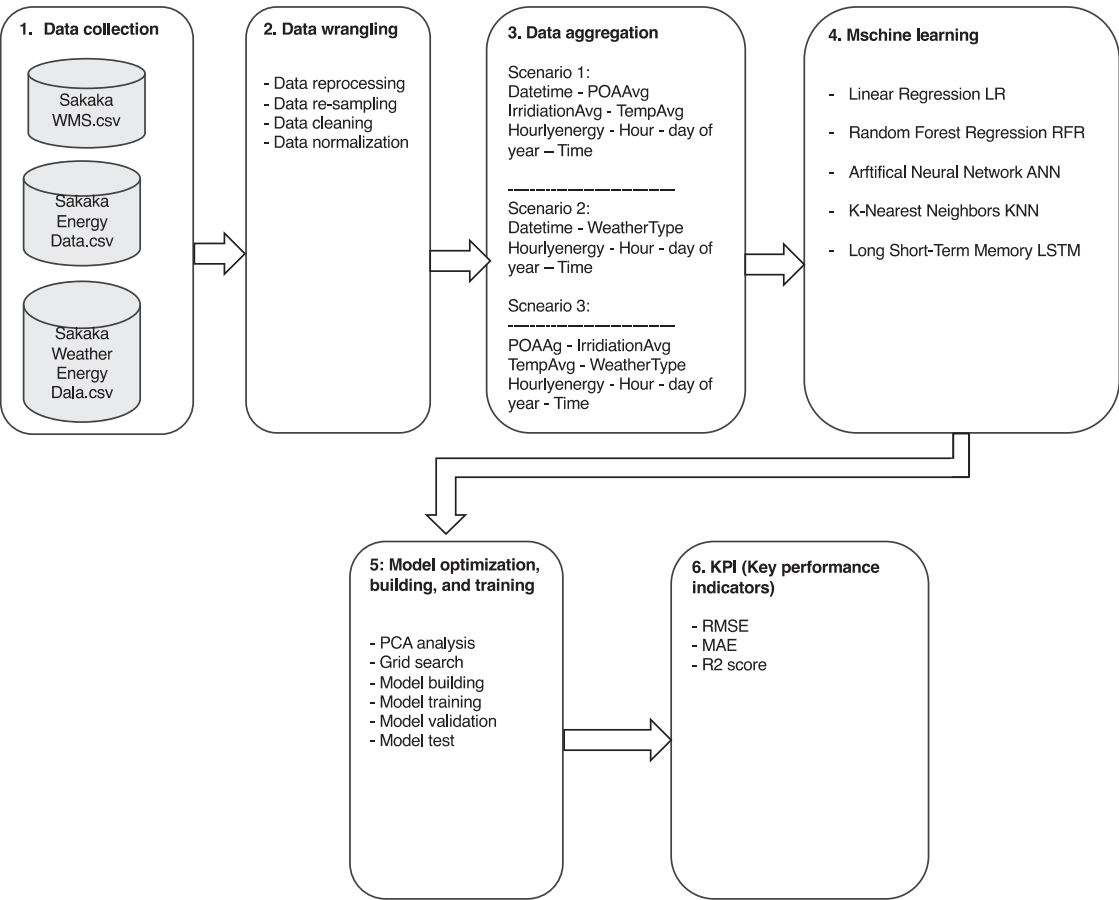


Fig. 6. Methodology overview.

4.3. Scenario definition

Three different scenarios are defined according to the variables used for the prediction, as detailed in Table 6. These scenarios were designed to analyze the impact of training features, training data sequence, and data quality on the accuracy of prediction. The key difference lies in the

features used to train the models.

- Scenario 1 (Solar data): In this scenario, solar irradiation and solar power data for two years were. Features include POA Average, Irradiation Average, and Temperature Average. This scenario was

Table 6
Different scenarios for merging data sources.

Datasets	Scenarios		
	1: Solar data	2: Meteo	3: Hybrid
Energy.csv	✓	✓	✓
WMS.csv	✓	—	✓
Weather.csv	—	—	✓
Samples	18,263	8679	8679
Features	DateTime POAAvg IrradiationAvg TempAvg Hourlyenergy Hour day of year Time	DateTime WeatherType Hourlyenergy Hour day of year Time	DateTime POAAvg IrradiationAvg TempAvg WeatherType Hourlyenergy Hour day of year Time

designed to assess the performance of machine learning models when trained solely on solar data.

- Scenario 2 (Meteo data): In this scenario, the weather data are considered. This scenario helps assess the impact of including weather information such as meteo type on solar power prediction accuracy.
- Scenario 3 (hybrid): It combines historical solar data and meteorological data. Features include historical solar data (from scenario 1) and weather data (weather type from scenario 2). This scenario aimed to assess whether combining both types of data would improve prediction accuracy compared to using either dataset individually.

These scenarios involve different combinations of features such as POAAvg (Plane of Array Average), IrradiationAvg (Irradiation Average), TempAvg (Temperature Average), WeatherType, HourlyEnergy, Hour, Day of Year, and Time. Note that to improve the accuracy in predicting the power output of solar panels, the time, day of the year, and hour columns are generated from the DateTime variable and then are appended to the merged dataset. These scenarios provide flexibility in exploring different combinations of features to obtain a more accurate prediction.

4.4. Model optimization, building, and training

The study employs several machine learning algorithms for solar prediction, including linear regression, random forest regression (RFR), artificial neural networks (ANN), closest neighbors, KNN, and long-short-term memory (LSTM). Before building the models, PCA (Principal Component Analysis) analysis and grid search are performed to optimize the model hyperparameters. This helps to improve the model’s performance and accuracy.

For ANN and LSTM models, a static network is adopted (Liu et al., 2017b). The neurons in the input and output layers are automatically determined by the dimensions of the input and output vectors. The weight and threshold values are randomly initialized. Regarding the number of hidden layers, a grid search is performed with cross-validation of 10 (CV=10) is used to determine the specific desired number of 2. The number of epochs and batch LSTM are empirically determined. The Nadam optimizer is used for both the ANN and the LSTM models. It is a computationally efficient optimization solver designed for neural network algorithms. It is specifically well-suited for handling large-scale problems characterized by a substantial amount of data or parameters. Additionally, Nadam’s memory requirements are minimal, making it an ideal choice for optimizing neural networks with limited computational resources. It combines the concepts of Root Mean Square Propagation (RMSprop) and Stochastic Gradient Descent with Momentum (SGDM). By merging the strengths of these two optimization

Table 7
Hyperparameters of algorithms.

Algorithm	Parameters
Random Forest Regression (RFR)	Number of trees: 150 Min samples split: 2 Min samples leaf: 1 Random state: 100
K Nearest Neighbors (KNN)	Number of neighbors (k): 5
Artificial Neural Network (ANN)	Number of hidden layers: 2 Epochs: 50 Activation function: 'relu' Optimizer: Nadam Learning rate = 0.001
Long Short-Term Memory	Number of hidden layers: 2 Batch: 32 Epochs: 50 Activation function: 'relu' Optimizer: Nadam Learning rate: 0.001

methods, Nadam aims to provide an efficient and effective approach to training neural networks.

For the KNN model (Yakowitz, 1987), the model assumes that similar weather conditions could result in a similar generation of photovoltaic power. When determining the neighbor number K, a predetermined range is established between 0 and 20 for the grid search algorithm with a cross-validation of 5 (CV=5). A value of K=5 is determined as the best value that leads to the best error performance.

RFR is an ensemble learning method that builds multiple decision trees during training and outputs the average prediction of the individual trees (Lahouar et al., 2017). In the training process, each tree is built using a random subset of the training data and a random subset of the features. This randomness helps to decorrelate the trees, reducing the risk of overfitting. During the prediction, each tree in the forest independently produces a prediction and the final prediction is the average of these individual predictions. RFR is robust to outliers, can handle non-linear relationships between features and the target variable, and implicitly performs feature selection by selecting the most key features for splitting nodes in the decision trees. Additionally, RFR requires minimal hyperparameter tuning and is parallelizable, making it efficient for training on large datasets. In our study, the number of decision trees is repeatedly tested from 50 to 1000 trees. A number of 150 leads to the best error performance. The number of 'min samples leaf' is set to 1, and the number of 'min samples split' is set to 2. In Table 7, the optimized hyperparameters for each learning algorithm used in the experiments are presented.

4.5. Validation and testing of the model

The trained models are validated and tested using key performance indicators (KPIs) such as the root mean square error (RMSE), the mean absolute error (MAE) and the score (R²).

MAE (Mean Absolute Error): MAE measures the average absolute difference between the predicted values, $\hat{y}_i$, and the actual values, y_i , without considering the direction of the error. It indicates the average magnitude of the errors. The formula for calculating MAE is as follows:

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i|$$

where n is the number of samples. Lower values indicate better performance.

RMSE (Root Mean Square Error): RMSE calculates the square root of the average of the squared differences between the predicted values, $\hat{y}_i$, and the actual values, y_i . It considers both the magnitude and direction of the errors. The formula to calculate RMSE is as follows:

Table 8

Experimental results of different machine learning models.

Model	Scenario	MAE	RMSE	R^2 Score
Linear Regression	Scenario 1	0.46	0.61	37.0 %
	Scenario 2	0.86	0.97	N/A
	Scenario 3	0.37	0.26	63.08 %
RFR	Scenario 1	0.06	0.15	97.73 %
	Scenario 2	0.05	0.11	98.55 %
	Scenario 3	0.03	0.09	99.05 %
KNN	Scenario 1	0.08	0.20	95.52 %
	Scenario 2	0.07	0.15	97.38 %
	Scenario 3	0.05	0.15	97.49 %
ANN	Scenario 1	0.08	0.19	95.90 %
	Scenario 2	0.10	0.16	97.03 %
	Scenario 3	0.05	0.11	98.71 %
LSTM	Scenario 1	0.16	0.32	88.08 %
	Scenario 2	0.15	0.28	90.86 %
	Scenario 3	0.09	0.19	95.72 %

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2}$$

R^2 Score (Coefficient of Determination).: The R^2 score represents the proportion of variance in the dependent variable that can be explained by the independent variable(s). It ranges from 0 to 1, where a value closer to 1 indicates a better fit. The formula to calculate the (R^2) score is as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

where $\bar{y}$ is the mean of the actual values. An R^2 score of N/A suggests that the model did not learn from the data.

5. Experimental results

In this section, the experimental results are presented and analyzed to evaluate the different learning models. We employ the Python programming language along with various integrated libraries, notably TensorFlow, pandas (for Data Analysis), matplotlib (for plotting), scikit-learn (for machine learning), NumPy (for numerical computing) and seaborn (for data visualization). Table 8 shows the different results obtained for the three scenarios.

The Linear Regression model produced high error values (MAE: 0.46, RMSE: 0.61) and a low R^2 score (37.32 %) for all scenarios. This is expected and is due to the nonlinear nature of the problem. The RFR model outperformed all other learning models, and this is for all scenarios. An MAE of 0.03, an RMSE of 0.09, and a R^2 score of 99.05 % were obtained for scenario 3. The model also performed well in scenarios 1 and 2. The KNN model demonstrated low error values for both scenario 1 and

scenario 2, with lower errors (MAE: 0.07, RMSE: 0.15) and a higher R^2 score (97.38 %).

The ANN model had low error values (MAE: 0.08, RMSE: 0.19) and a good R^2 score (95.90 %) in scenario 1 and performed slightly worse in scenario 2, with higher errors (MAE: 0.10, RMSE: 0.16, and a slightly lower R^2 score (97.03 %). However, in scenario 3, the model performed better than in the other scenarios, with lower errors (MAE: 0.05, RMSE: 0.11) and a high R^2 score (98.71 %).

The LSTM model had moderate error values (MAE: 0.16, RMSE: 0.32) and a lower R^2 score (88.08 %) compared to other models, with the best performance in scenario 3 (MAE: 0.09, RMSE: 0.19, R^2 score (95.72 %). In general, the Random Forest Regressor (RFR) provides the best performance for all three scenarios, consistently producing the lowest error values (MAE and RMSE) and the highest R^2 scores. KNN and ANN models also showed good performance, while the linear regression and LSTM models had higher error values and lower R^2 scores, which makes them less suitable for this solar energy prediction task.

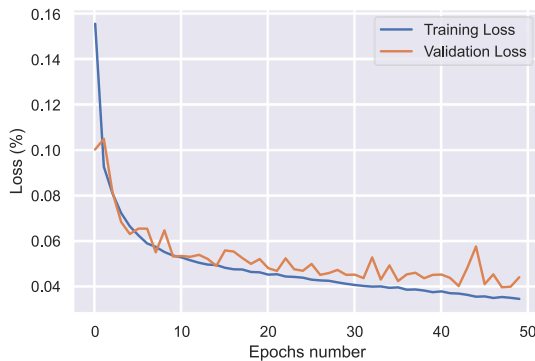
The ANN and LSTM models were trained for several epochs and quickly converged without overfitting the training data (Fig. 7). The performance of the ANN model was shown to be better than that of the LSTM model by overfitting the training data and allowing for an extensive learning process.

Fig. 8 shows the scatterplots of the observed energy power versus the predicted power obtained from all models for scenario 3. The comparison reveals that the RFR model has the highest variability with an R^2 score of 99 %. The ANN, LSTM, and KNN models had slightly lower variability, with R^2 scores of 0.98 %, 96 %, and 97.5 % respectively, as illustrated in Fig. 8. Furthermore, we show how KNN is too sensitive to outliers. The RFR model can handle outliers to some extent because of their ensemble nature and thus is less sensitive to individual data points. The ANN and LSTM models are more robust and can handle outliers better than other learning models.

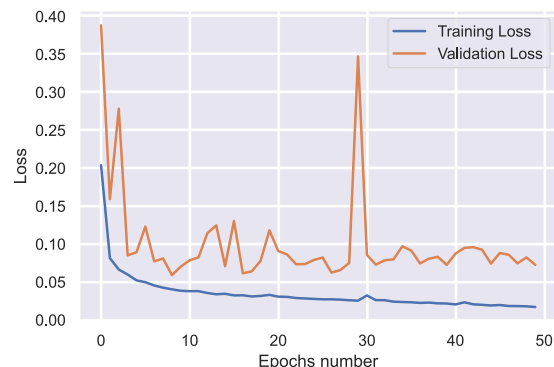
5.1. Runtime analysis

Fig. 9 shows the runtime performance of the five algorithms. LR and KNN have the lowest runtimes. However, linear models fit poorly the data given the nonlinear nature of the problem. RFR has slightly higher runtimes compared to LR and KNN but remains fast. It presents consistent performance with runtimes of the order of milliseconds. The ability of RFR to handle complex relationships and provide accurate predictions with fast computations makes it a promising choice for solar energy prediction tasks.

On the other hand, ANN and LSTM have significantly higher runtimes compared to the other algorithms. Both ANN and LSTM involve more complex computations due to their deep learning architectures, which require more time to process and train the models. The runtime of ANN ranges from seconds to a few minutes, while the runtime of LSTM



(a) ANN model convergence



(b) LSTM model convergence

Fig. 7. Model convergence: ANN and LSTM models.

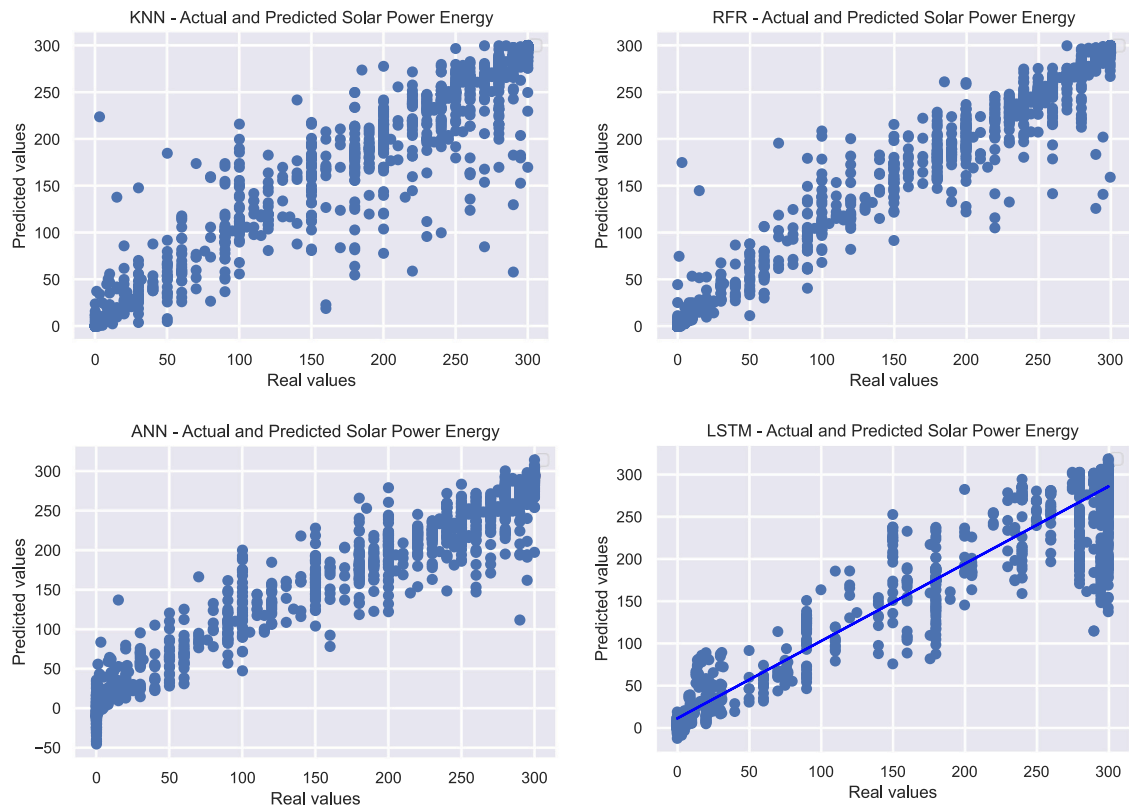


Fig. 8. Scatter plots of the observed against forecasted power for: KNN model, RFR model, ANN model, LSTM model.

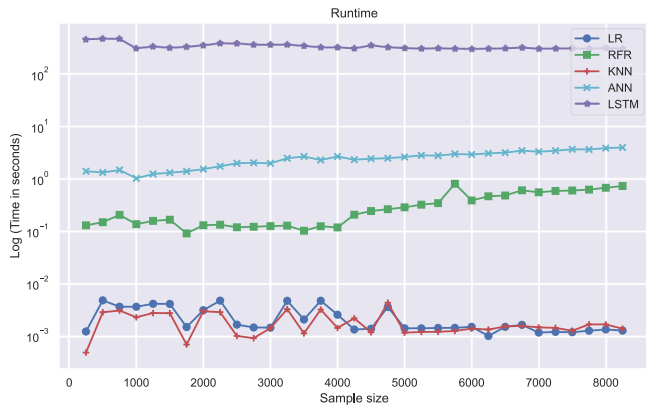


Fig. 9. Runtime analysis.

exceeds minutes in some cases. Considering the runtime Analysis, RFR is the most efficient algorithm in computational speed and prediction accuracy.

In conclusion, the main reasons for RFR’s outperformance are: (1) handling nonlinearity, (2) overfitting robustness, and (3) handling outliers and missing data. Artificial Neural Networks are considered even more robust in handling outliers compared to other models; however, the tuning of hyperparameters is a complex and challenging task. More research should be done on the tuning of hyperparameters of artificial neural networks to better handle other dynamic models.

6. Conclusion

Forecasting the energy yield of photovoltaic systems (PV) is an effective and economical way of managing the grid and is a key factor in ensuring the dependability of the system when photovoltaic systems are

widely used. In this study, a comparative analysis of machine learning models is presented to forecast solar energy generation. The analysis was carried out using three years of meteorological and solar data obtained from a large-scale industrial project. The proposed methodology evaluated the performance of several machine learning models, namely linear regression, random forest regression, K-nearest neighbors, artificial neural networks, and long short-term memory. Common performance metrics are used for the evaluation as follows: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Coefficient of Determination (R^2). Different scenarios have been defined to measure the effect of training features, training data sequence, and data quality on prediction accuracy.

The experimental results revealed that the Random Forest Regression model outperformed other learning models for all scenarios. It produced an MAE error of 0.06 % when using solar irradiance data and 0.03 % when using the hybrid regime that combines solar data with meteorological data. Although artificial neural networks exhibited strong performance, they required more comprehensive hyperparameter adjustments and a greater demand for computational resources. Moreover, combining meteorological data with solar irradiance data further improved the prediction accuracy of all models and scenarios by at least 0.03 %.

Exploring hybrid models that combine multiple techniques would potentially lead to better prediction accuracy. Real-time forecasting is another key point to consider in the future. Refine the models for real-time or short-term forecasting (minutes to hours) to accommodate rapidly changing weather conditions and improve grid stability. Furthermore, the use of more advanced deep learning architectures, such as convolutional neural networks (CNNs) would be considered. This might be particularly adept at handling complex spatial-temporal relationships in solar energy data.

CRediT authorship contribution statement

Alain, Khaled Chaaban: Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Data curation, Conceptualization. **Najd Alfadl:** Writing – original draft, Funding acquisition, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The data that has been used is confidential.

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Author agreement statement

We the undersigned declare that this manuscript is original, has not been published before and is not currently being considered for publication elsewhere. We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us. We understand that the Corresponding Author is the sole contact for the Editorial process. He/she is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs

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